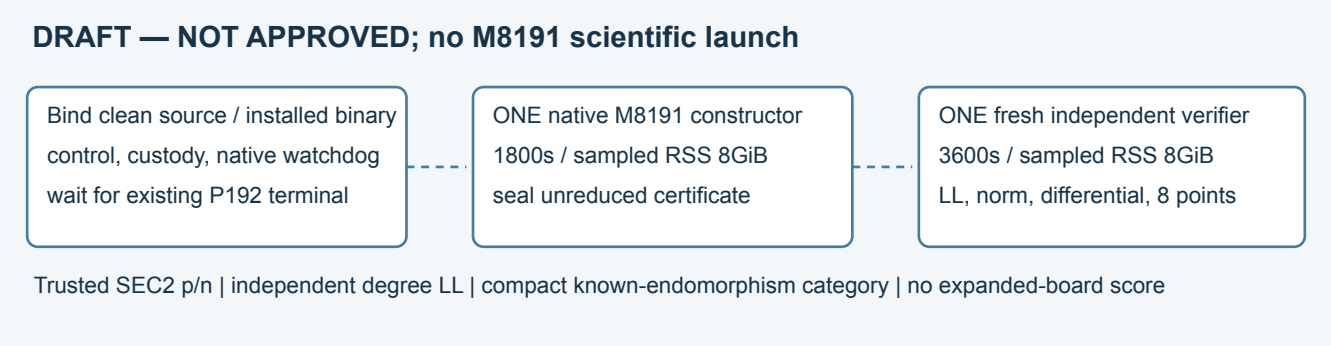


M8191 compact-CM protocol draft — not approved for execution

Goal GOAL-ISOU-f30a07. Proposed experiment EXP-ISOU-feb102 and hypothesis H-ISOU-d1a4a5; identifiers must be checked again when filed. This is preparation only, with `execution_authorized: false`. No experiment approval, trial launch, candidate-degree result or worldwide claim is created. The current source is still being implemented and tested; no clean source/binary hash is guessed.

Exact uncertainty and scope

Does the reviewed native implementation construct a compact secp256k1 endomorphism expression of geometric prime degree ell=2^8191−1 and pass a fresh, independent native verification within the declared machine protections? The representation is **compact known-endomorphism self-map**, separate from the expanded-map challenge board and season. The source and endpoint are the exact same pinned SEC2 secp256k1 model, including p,a=0,b=7,n,h=1,G and nontrivial beta. SEC2 primality of p,n is explicitly **trusted standard, not self-contained certificates**. Degree primality is independently replayed Lucas–Lehmer under Conrad Theorem2.2, with odd prime exponent and exactly8189 transitions; Gross-specific GCD diagnostics are optional.

The constructor uses BigUint arithmetic, Mersenne-fold LL reduction and Jacobian point formulas. The verifier uses the repository Int with general modular reduction, independent Cornacchia replay, fixed-limb affine arithmetic and both dual compositions. These independence claims must be checked against the committed implementation, not inferred from module names. Exact unreduced signed coefficients and their norm, the beta/lambda full-point pairing, nonzero differential and subgroup scalar/inverse, integer polynomial expansion, and the root IDC1 evaluated supplement are separately checked. The IDC1 supplement alone establishes neither an integer identity nor geometric degree.

Frozen input and build gate

Before approval, bind the clean source commit and source-tree receipt, exact installed executable path and SHA256, binary custody, compiler/build environment, exact focused-control receipt, underlying source/catalogue hashes, and the machine-protection entry point/hash. The CLI must be usable without Cargo at run time. Source changes after binding require a new source/binary amendment before execution. Preserve the implementation review and its corrections in the input set: docs/reports/isogeny-cm-norm-design-20261016/independent-review.md and DEC-20261016-3fb077.

The latest, still-uncommitted compact implementation now routes public construct/verify through the existing native supervisor::run, with public --timeout default/caps1800/3600. The hidden single worker waits for a positive RSS acknowledgement before arithmetic and signals completion; sampling is25ms with a100ms transient monitor grace. Approval binds the clean committed source/binary and passing focused resource-control receipt; reading uncommitted source is not that receipt. All native.json, native.stderr.txt, native.pid, native.receipt.json, resources-ready and completion markers, wrapper records and artifact subdirectories are retained. The verification consumer bounds certificate input at2MiB and hashes the exact same byte snapshot it parses. No Python mathematical or experiment implementation is introduced by this draft.

Fixed stages, samples and controls

One fresh constructor run RUN-ISOU-bdbf5d, then one fresh independent verification RUN-ISOU-2b2372 only after constructor exit0/statusCONSTRUCTED and a sealed certificate. All filesystem arguments are absolute. Neither existing output may be replaced, and no retry/adaptive exponent is included. Constructor success alone is not verification. Fresh verify binds the precise certificate hash and must not reuse a producer verification verdict.

The point set is exactly8 deterministic public inputs: O,G,−G,[2]G,[3]G,[17]G,[101]G,[65537]G. They are fixed known-answer inputs, not unseen random targets. Full point coordinates, infinity handling, geometric and subgroup evaluator agreement, both dual compositions, curve membership and group-law compatibility are checked. Point agreement does not prove degree. No expanded kernel or rational-map polynomial is promised by this category.

The committed focused implementation controls cover exponent3/degree7, exponent5/31, exponent7/127; prime exponent11 with composite M11=2047; composite exponent9; and declared handling of exponent2. They must also cover beta=1, wrong beta/lambda pairing, norm/source/identity/hash/recurrence mutations, and the degree alias alpha+[n]. The alias keeps rational-point outputs but must fail the original norm-degree certificate. An identity coefficient changed by2^61−1 must fail the exact integer checker even if finite-field evaluation cannot distinguish it. The protocol binds those receipts; it does not silently rerun an unbounded control grid during the scientific instance.

Machine protection and full cost accounting

Constructor wall watchdog1800s, fresh verify3600s, sampled worker RSS8GiB per phase, certificate input2MiB, post-run serialized output admission limit256MiB, at most one scientific process for this goal, and no unaccounted worker. Native worker sampling is25ms; preserve the actual receipt and label it sampled rather than true continuous peak. The supervisor is a separate process; its own resource use is reported where measured rather than claiming an unmeasured total-tree peak. Monitoring unavailable beyond the native100ms grace, a lost process or absent terminal receipt makes the attempt INCOMPLETE. Brief unavailable samples inside that recorded grace remain explicit in the receipt. Only one computational worker is authorized. If an implementation adds multiple workers, a new resource-binding amendment must define and enforce combined RSS before launch. Watchdog limits protect this host; they do not cap campaign spending or prove mathematical impossibility. Retain stdout/stderr and partial outputs when any limit trips.

Report total constructor and independent-verifier wall/CPU/RSS/bytes separately, including all LL, Cornacchia, coefficient, identity and point work inside their phase totals. If individual internal timings are unavailable, report them as unavailable and do not invent a component breakdown. Charge failed attempts as well. Retain command/cwd, UTC start/end, host/OS/CPU, source/binary/input/output hashes, status/exit/signal, watchdog sample receipt and directory manifest. Native elapsed_ms_operational_only is descriptive; the native supervised terminal receipt supplies the worker phase boundary; record outer command wall/CPU separately where available. No performance comparison or speedup is predeclared.

Terminal states and next decisions

PASS requires constructor CONSTRUCTED and fresh verifier exit0/statusVERIFIED_COMPACT, degree_bits8191 and exact degree2^8191−1, every required check passing, complete control/build/custody/phase receipts, matching hashes, trusted-standard domain status explicit, and no expanded-map score. Archive the producer outputs before an independent implementation/integrity review; only subsequent Coordinator review can consider compact-category promotion.

A completed algebraic mismatch with a concrete failing identity is an implementation failure at this instance. Timeout/RSS/size/missing-telemetry/crash/lost process is a censored or infrastructure observation, with no degree admission and no nonexistence inference. Preserve all bytes, return the exact stopped phase, and rerank before any successor. No automatic rerun is authorized.

The goal concurrency is one: the existing P-192/50341 job must reach its own terminal receipt before this compact trial is launched. Its T3150 retry budget is exhausted; do not redispatch, reclaim or restart it. This protocol cannot supply a route around that boundary. An observation/archive task after its terminal state is separate postprocessing and starts no scientific process.

Approval and dispatch conversion

Fill every binding, verify the existing native launcher under its actual resource controls, file a complete frozen experiment/hypothesis/plan and Coordinator decision under standing authorization, publish the input snapshot and handoff/queue, and check readiness. The current draft is not a trial-plan.json, has no approval and cannot be selected as runnable. The exact draft command vectors are recorded in commands.draft.json to make the final binding mechanical. No target command has been run while preparing it.

Conditional P-192 next-action ranking

This is a decision rule awaiting the actual terminal receipt of RUN-ISOU-f9546d, not an empirical result or a new launch. Do not restart the existing replay PID or T3150. Keep all original failure, control, constructor, map and replay records intact.

1. First archive the terminal constructor/replay/resource receipts under a genuine postprocessing-only scope. Record construction and replay separately; completion/exit/signal, sampled RSS availability and peak, map and receipt hashes, one-eigenline scope, and total charged wall/CPU. A missing receipt is INCOMPLETE, not a zero cost or failure theorem. The current map SHA supplied by the Executor is82e3951488c793c5901ecf1dfeb3ab072b7cda16730899330df7caa9832a170c; independently bind it at archive time.
2. If replay passes, independent integrity review comes before a P-192 baseline promotion. If it times out or fails, preserve its exact phase and keep degree admission unestablished. No larger degree is dispatched merely because construction completed.
3. Rank same-artifact replay profiling/optimization ahead of constructing a larger expanded map whenever terminal phase accounting shows replay is the dominant measured cost or its limit censored admission. Reuse the exact sealed50341 map; do not regenerate it. Do not state which internal routine dominates without profile evidence.
4. Inspect/profile the actual generated replay engine closure: tools/isogeny-cli/build.rs selects fixed-limb verification_field, research/.../verification_poly.rs and verification_kernel.rs, tools/.../verification_map.rs and replay.rs. The active polynomial implementation already has Karatsuba and fast remainder paths. A generic schoolbook implementation elsewhere is not evidence about this binary. Candidate discriminating measurements isolate squarefreeness/GCD, division-polynomial torsion, subgroup modular composition, exact rational-map substitution, and point transports while preserving every exact algebraic check.
5. After a frozen profile identifies the dominant stage, propose one semantics-preserving optimization with matched same-map baseline/candidate binary receipts and meaningful mutation controls. Use A/A pairing when asserting a timing advantage, charge warmup/setup, and preserve implementation independence. If the old baseline is censored, report the bound and censoring; do not manufacture a completed runtime or a point speedup ratio. A candidate that finishes must still pass every algebraic and adversarial check.
6. If replay passes comfortably and accounting does not justify optimization, rerank a single next degree from the predeclared supported constructor/eigenline grid using projected full constructor+replay memory/time/bytes, then freeze a fresh exact candidate. Do not pick an adaptive degree during this live run or extrapolate a worldwide record.

At present the next P-192 scientific action is unselected because terminal phase cost is still missing. This is not a held research goal: GOAL-ISOU-f30a07 remains active, with the live run and separate compact native implementation progressing. No new mathematical status is recorded here.